



Modern CAD Design Plan & Work Tracker

Close the gap from FreeCAD toward SolidWorks and Fusion 360, starting on the FreeCAD foundation and shipping first on Apple Silicon.

Repository **brucecodes2023/FreeCAD_BT**
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Branch **cursor/modern-cad-phase1-953c**

Updated

HOW TO USE THIS DOCUMENT

This is the working design record for the fork. IDs are stable. When work lands, flip the status tag, name the files, and add a line to the work log at the end. The original Cloud Agent scoping note (`docs/macos-native-ui-scope.tex` on branch `cursor/setup-cloud-agent-env-d2a0`) covered A/B/C only. This file keeps those IDs and adds shell (D), look (E), modeling UX (F), and simulation (G).

GitHub does not render `.tex`. Keep the Markdown edition open in a browser tab while we work: `Documents/README.md` (also linked from repo-root `PLAN.md`). The PDF in this folder is the typeset copy; CI rebuilds it as the **Design PDF** Action artifact when this source changes.

Constraints that still hold

- Source-only until a Mac checkout builds and launches the GUI. Do not treat a Linux VM as a visual preview.
- Work from FreeCAD's existing extension points (preference packs, YAML/QSS, workbench `ToolBarItem` trees, Elmer equation objects). Do not start with a Metal rewrite or a ribbon-only skin.
- Effort ratings describe technical scope, not calendar time.

Product thesis

SolidWorks is the target *feel* (FeatureManager, CommandManager-like strip, familiar mouse, guided simulation). Fusion is the target *shell* (ribbon while modeling, dashboard on launch). FreeCAD is the *foundation* (OCCT kernel, Body/Tip history, workbenches, CalculiX + Elmer). The gap is closed in this order, because that is the order the codebase actually supports:

1. **Shell** — Home dashboard + Fusion-style ribbon over existing commands.
2. **Look** — YAML tokens, QSS, icons. Not a C++ widget rewrite.

3. **Modeling UX** — Body/Tip, naming, rollback, sketch-in-tree.
4. **Mac smoothness** — Qt/OpenGL mitigation now; Metal later.
5. **Simulation** — guided CalculiX plus Elmer first-principles as WIP.

Keep the native macOS menu bar. The ribbon is the command strip, not a replacement for File / Edit / View.

Status tracker

SHIPPED AND NATIVE MAC

ID	Item	Status	Notes
A1	SolidWorks nav default	Done	New profiles only; existing <code>User.cfg</code> unchanged.
A2	Modern CAD behavior pack	Done	Nav, cube, combo view, ribbon flag.
A3	View cube de- faults	Done	Top-right, size 132.
A4	First-start on- boarding	Done	Applies Modern CAD pack once.
A5	Fusion 360 nav class	Done	Additive; MMB pan / Shift+MMB orbit. Default stays SolidWorks.
A6	Fusion-style ribbon	Partial	QAT, overflow, captions, sketch contextual tabs. Mac QA remaining.
B0	Traffic lights already native	Done	Finding; no code move.
B1	Unified title + toolbar	Done	<code>setUnifiedTitleAndToolBarOnMac(true)</code> .
B2	Native dock/- toolbar chrome	Done	Cocoa chrome; <code>macos-native-chrome.qss</code> .
B3	NSWindow full- size content	Later	New <code>.mm</code> shim.
B4	Bundle id + dark mode	Done	<code>org.freecad.FreeCAD</code> ; Aqua appearance off.
C1	arm64 + ma- cOS 11 pin	Done	CMake helpers + conda-macos preset.
C2	Optional LTO / apple-m1	Done	Off by default.
C3	Enable TBB	Done	<code>HAVE_TBB</code> + <code>ImportOCAF</code> .
C4	Defer Init- Gui.py	Later	Startup ordering risk.
C5	Metal / Qt RHI viewport	Later	Real M5 fix. Near-term: vsync + software GL.

SHELL, LOOK, MODELING, SIMULATION

ID	Item	Status	Notes
D1	Home dashboard	Partial	Metrics, tips, command search, projects, new-file cards.
D2	Project folders	Partial	Create + list; open file from folder.
D3	Dashboard health widgets	Partial	Drawing + Continue; recompute errors; GPU warning; search.
E1	Modern CAD Light/-Dark tokens	Partial	Cool gray + SW blue; iterate on Mac.
E2	Denser Feature tree / icons	Partial	Pack: 32px toolbars, tree icon 24 / indent 10.
E3	Viewport / PD pre-view colors	Partial	Gradient off; dress-up preview blue-gray.
E4	Custom sketch/feature icons	Later	Optional icon set.
F1	Rollback bar	Later	Body/Tip history UX.
F2	Topological naming UX	Later	Kernel work, not chrome.
F3	Sketch-in-tree FeatureManager	Later	Single tree like SW.
F4	In-context assembly edits	Later	Beyond Ondsels joints.
G1	First Principles study stub	WIP	Elmer elasticity + heat; labeled WIP.
G2	Guided equation wizard	Partial	Mesh / material / BC checklist before solve.
G3	CalculiX study pre-sets	Partial	Static + thermal wrappers; no auto-mesh.
G4	Assembly contact FEA	Later	Large gap vs SW Simulation.

Workstream A CAD-familiar defaults

Goal: a first launch that feels like SolidWorks / Fusion without touching preferences. Vehicle: built-in preference pack plus shipped fallbacks.

A1 — SolidWorks navigation

Navigation styles live under `src/Gui/Navigation/`. `SolidWorksNavigationStyle` already existed. New profiles now fall back to it, with turntable orbit and object-center rotation (`DefaultNavigationStyleName`, `DefaultOrbitStyle`, `DefaultRotationMode` in `NavigationStyle.h`). Existing `User.cfg` is not migrated.

A2 — Modern CAD behavior pack

Pack at `src/Gui/PreferencePacks/Modern CAD/`: SolidWorks nav, cube, combo-view layout, `UseRibbon=1`. Type is Behavior, not Theme — apply on top of Light/Dark.

A3 — View cube

NaviCube top-right, size 132, rotate-to-nearest. Configurable via `BaseApp/Preferences/Nav-iCube`.

A4 — First-start onboarding

Start workbench `FirstStart2024` flow applies the Modern CAD pack once (`ModernCADDefaultsApplied`), skipped in GUI tests. Copy mentions SW/Fusion nav and the ribbon/dashboard.

A5 — Fusion 360 navigation class

`Gui::Fusion360NavigationStyle` is registered. Fusion defaults: MMB pan, Shift+MMB orbit, scroll zoom. New profiles still use SolidWorks. Pick Fusion 360 in Preferences → Display → Navigation.

A6 — Fusion-style ribbon

Originally deferred. Product decision reversed: ribbon *replaces* classic toolbars while a document is open.

Shipped

- `Gui::RibbonBar / RibbonManager` consume the workbench `ToolBarItem` tree.
- Tabs = toolbars; groups split on separators; workbench combo on the tab row.
- Hidden when the active MDI view is `StartView`.
- Classic `QToolBars` hidden while ribbon is on (including after sketch `setState`).
- Preference `BaseApp/Preferences/MainWindow/UseRibbon` (default on; off in GUI tests).
- Ribbon sits above the MDI area, not inside the 22 px unified title bar.
- Quick Access strip: New, Save, Undo, Redo.
- Per-page overflow chevron (») when groups do not fit.
- Fusion-sized captions (toolbar icon size from prefs, text under icon).
- Contextual tabs: `Unavailable` (sketch edit) toolbars shown only while a sketch is in edit.

Still to polish on Mac: height vs traffic lights, overflow at laptop widths, optional QAT customization.

Workstream D Home dashboard

The landing page must *not* show the ribbon. It is a dashboard.

D1 — Dashboard chrome

`StartGui::StartView` documents page is now Home: title, three metric cards (recent files, project count, graphics backend), then projects, new-file cards, recent files, examples. Metrics are placeholders for later health/solver widgets.

D2 — Projects

`Start::ProjectsModel` stores folder paths under `BaseApp/Preferences/Mod/Start/Projects`. New Project picks a parent folder, creates a named directory and a `.freecad-project` JSON marker, then lists the project. Clicking opens a file dialog in that folder. Not a SolidWorks assembly project yet — a workspace folder.

D3 — Dashboard widgets (partial)

- Templates: Part Design Body, Assembly, Drawing (TechDraw), plus Continue last file.
- Document health: recompute error count across open documents.
- In-page command search (`Gui::CommandCompleter`).
- Units / navigation style on a metric card.
- What's-new / first-run tips (dismissible).
- Mac GPU warning when the OpenGL renderer string looks like Metal 90.x.

Pinned folders and sketch-solver health remain later.

Workstream E SolidWorks-like facelift

Theming is YAML tokens + `FreeCAD.qss` (`StyleParameters::ParameterManager`), not a C++ skin.

E1 — Modern CAD Light / Dark

New packs and YAML:

- Light: chrome #F5F5F7, accent #0066B3, flat padding, light-gray 3D background.
- Dark: graphite #2D2D30, brighter blue accent.

Apply from Preferences → Preference Packs, on top of the Modern CAD behavior pack. Visual QA waits for a Mac launch.

E2–E4 — Density, viewport, icons

Packs set toolbar icons to 32 px and denser Feature tree (icon 24, indent 10, item spacing 1). Viewport gradient stays off. PartDesign dress-up preview is blue-gray instead of magenta (`StyleParameters.h`). Custom SW-like glyphs (E4) remain later.

Workstream F Modeling kernel UX

This is the real SolidWorks gap. Chrome cannot fake it.

Today. PartDesign is Body + Tip + history. Pad/Pocket/Fillet go through ProfileBased / DressUp. Recompute is `App::Document::recompute()` topo-sort. Assembly is `AssemblyObject` + `OndselSolver` joints (kinematic), not SW mates.

Gaps vs SolidWorks

- F1 — no real rollback bar on the history tip.
- F2 — topological naming is still fragile after face edits.
- F3 — sketch is a separate object; no single FeatureManager tree.
- F4 — no in-context assembly edits / SW-style mates driving geometry.

Do not schedule F before D/E are usable on a Mac. When F starts, work in `src/Mod/PartDesign/` and `src/App/Document.cpp`, not in QSS.

Workstream G Simulation / first principles

Today. `FemAnalysis` plus external solvers. CalculiX is the structural workhorse. Elmer is the continuum-PDE path (elasticity, heat, flow, EM equation objects under `src/Mod/Fem/`).

G1 — First Principles Study

Command `FEM_FirstPrinciplesStudy` (Model + Solve toolbars/menus): creates or reuses an analysis, adds Elmer, attaches elasticity + heat equations, shows a one-time WIP dialog. Does *not* replace CalculiX, add contact, or invent a SW Simulation study tree.

G2–G4 — Wizard, CalculiX presets, contact

`FEM_StudyGuidedWizard` is a mesh / material / BC checklist (no new solver).

`FEM_CalculiXStaticStudy` and `FEM_CalculiXThermalStudy` create analysis + CalculiX + empty material. Assembly-wide contact (G4) remains a large gap, not a stub.

Workstream B macOS-native chrome

B0 — Traffic lights already native

`Gui::MainWindow` is a plain `QMainWindow`. Qt/Cocoa draws the standard frame; red/amber/green already sit top-left. No code move.

B1 — Unified title + toolbar

`setUnifiedTitleAndToolBarOnMac(true)` in the `MainWindow` constructor. The ribbon is *not* placed in that unified bar (too tall); it is a widget above the MDI area.

B2 — Native dock & floating toolbars

Floating toolbars keep native Cocoa chrome. Overlay `src/Gui/Stylesheets/macos-native-chrome.qss` is appended on `Q_OS_MAC`.

B3 — NSWindow full-size content

Content under the title bar / large-title behavior needs a new Objective-C++ `.mm` shim. Precedent exists (QuickLook, 3Dconnexion) but no window-chrome Cocoa code yet.

B4 — Bundle identity & appearance

`CFBundleIdentifier=org.freecad.FreeCAD`, `NSRequiresAquaSystemAppearance=false`. Signing/notarization is a release concern, not this branch.

Workstream C Apple Silicon & M5 smoothness

Why M4/M5 feel laggy

The 3D view is OpenGL via Coin3D + `QOpenGLWidget` (`src/Gui/Quarter/QuarterWidget.cpp`, `src/Gui/View3DInventorViewer.cpp`). On macOS that OpenGL is Apple's Metal translation layer (OpenGL 2.1 Metal - 90.x). M5 GPUs are Metal-4 oriented; the translation path is the known hitch/crash source (upstream reports on M5 MacBook Air + macOS Tahoe, slow Sketcher recompute on M4). Safe mode does not fix it. Qt 6.8.3 has a related `QMacAccessibilityElement` use-after-free on Tahoe.

More OpenGL tweaks will not close this gap. C5 (Metal / Qt RHI) is the real fix.

C1 — Architecture pin

`CMAKE_OSX_ARCHITECTURES=arm64` when building on Apple Silicon; default
`CMAKE_OSX_DEPLOYMENT_TARGET=11.0`.

C2 — Optional LTO / `-mcpu=apple-m1`

`FREECAD_USE_LTO` and `FREECAD_APPLE_SILICON_TUNING` are off by default.

C3 — TBB

`find_package(TBB)` before compiler checks so `HAVE_TBB` in `src/config.h.cmake` lights up `ImportOCF` parallel paths.

C4 — Faster startup

All modules' `InitGui.py` still run eagerly (`src/Gui/FreeCADGuiInit.py`). Deferral has ordering risk.

C5 — Metal / Qt RHI

Near-term (shipped): `QSurfaceFormat::setSwapInterval(1)` on macOS; software OpenGL fallback tooltip updated for M-series. Prefer Qt newer than 6.8.3 on Tahoe when building. Do not treat nav cube, extra GL contexts, or QSS as the root cause.

Sequencing

Phase	Focus	Items
Phase 1	Familiar defaults	A1, A3, A2, A4 Done
Phase 2	Native Mac chrome + arm64	B1, B2, B4, C1, C2, C3 Done
Phase 3	Shell + look v0	A6, D1, D2, E1, G1 Partial / WIP
Phase 4	Shell/look polish (source in; Mac QA)	A6 polish, D3, E2–E3 Partial
Phase 5	Modeling UX	F1–F4
Phase 6	Simulation depth + Metal	G4, C5, B3, C4 (A5/G2/G3 started)

Phase 3 is the current working phase. Modeling kernel (F) and Metal (C5) wait until a Mac build has validated Phase 3 visually.

Validation plan

Change	How it is proven
A1–A4 defaults / pack	Fresh profile: SW orbit, cube top-right, Modern CAD applied once.
A6 ribbon	Document open → ribbon + QAT; Home → no ribbon; overflow »; sketch edit shows contextual tabs.
D1–D3 dashboard	Home shows metrics, tips, search, New Project, Drawing, Continue.
E1–E3 look	Modern CAD Light/Dark; 32px icons; flat viewport; PD dress-up blue-gray.
G1–G3 FEM	First Principles Elmer stub; Guided Wizard; CalculiX static/thermal presets.
A5 Fusion nav	Preferences lists Fusion 360; default remains SolidWorks.
B1/B2/B4 Mac chrome	Traffic lights top-left; unified toolbar; dark mode follows system.
C1/C3 build	Reports arm64; TBB found; ImportOCAF can use it.
C5 near-term	Vsync on; software GL option documented; M5 still expected to need Metal.
No regressions	FreeCAD CLI + C++ self-tests remain green. GUI tests force ribbon off.

Grounded file index

Concern	Key path(s)
This plan	Documents/modern-cad-design.tex
Navigation defaults	src/Gui/Navigation/NavigationStyle.h, View3DSettings.cpp
Modern CAD packs	src/Gui/PreferencePacks/Modern CAD/ and Light/Dark
Ribbon	src/Gui/RibbonBar.cpp, RibbonManager.cpp, Work- bench.cpp
Ribbon QSS	src/Gui/Stylesheets/fusion-ribbon.qss
Home dashboard	src/Mod/Start/Gui/StartView.cpp, ProjectsModel.cpp
Theme tokens	src/Gui/Stylesheets/parameters/Modern CAD Light.yaml
Main window chrome	src/Gui/MainWindow.cpp, ToolBarManager.cpp
macOS bundle	src/MacAppBundle/FreeCAD.app/Contents/Info.plist
arm64 / TBB / LTO	cMake/FreeCAD_Helpers/, SetupTBB.cmake
OpenGL / vsync	src/Gui/Application.cpp, Quarter/QuarterWidget.cpp
FEM first principles	src/Mod/Fem/femcommands/commands.py, femtools/s- tudy_presets.py
FEM wizard	src/Mod/Fem/femguiutils/study_wizard.py
Fusion 360 nav	src/Gui/Navigation/Fusion360NavigationStyle.cpp
PartDesign kernel	src/Mod/PartDesign/App/Body.h, src/App/Docu- ment.cpp
Assembly	src/Mod/Assembly/App/AssemblyObject.*
GUI startup	src/Gui/FreeCADGuiInit.py, StartupProcess.cpp

Compile and launch on a Mac

Build from source on Apple Silicon with pixi. Do not treat a Linux VM as a visual preview. Full copy-paste lives in Documents/README.md; this is the short form.

Prerequisites. Xcode Command Line Tools (`xcode-select -install`) and pixi (`curl -fsSL https://pixi.sh/install.sh | bash`). Homebrew Qt is not used. Plan on about 20 GB free (clone + pixi env + build/release objects). `install-release` is optional for a first launch. macOS 27 beta is newer than the Qt 6.8.3 pin — save the log if configure or the GUI fails.

```
git clone https://github.com/brucecodes2023/FreeCAD_BT.git
cd FreeCAD_BT
```

```
git checkout cursor/modern-cad-phase1-953c
pixi install
pixi run initialize
pixi run configure-release
pixi run build-release

export FREECAD_USER_HOME="$HOME/freecad-modern-cad-fresh"
mkdir -p "$FREECAD_USER_HOME"
pixi run freecad-release
```

Use a fresh FREECAD_USER_HOME so A1 defaults apply. After launch: Home has no ribbon; opening a document shows the ribbon and QAT; apply Modern CAD Light/Dark on top of the Modern CAD behavior pack; Fusion 360 nav is optional in Preferences. If the 3D view lags, enable software OpenGL and restart. C5 Metal stays later.

The pixi build does *not* install into /Applications. After build/release/bin/FreeCAD exists, pixi run macos-app writes ~/Applications/FreeCAD_BT.app (a launcher that still needs this checkout). pixi run macos-cleanup reclaims ccache and debug objects. The install is full FreeCAD (CAM, BIM, FEM, ...). Pixi is Apple Silicon only; Windows/Linux sources stay because they are the same CAD kernel.

Work log

Update this list every time the plan or the code moves. Newest first.

- **13 Aug 2026 — Remaining IDs (source).** A5 Fusion nav; A6 QAT / overflow / contextual tabs; D3 dashboard widgets; E2/E3 density + PD previews; G2 wizard; G3 CalculiX presets. Mac compile walkthrough added. F1–F4, C5 Metal, B3, C4, E4, G4 still later.
- **13 Aug 2026 — Documents/ revived.** Living LaTeX plan replaces the missing docs/macos-native-ui-scope.tex. Status tracker covers A–G.
- **13 Aug 2026 — Phase 3 v0.** Fusion ribbon (A6), Home dashboard + projects (D1/D2), Modern CAD Light/Dark (E1), FEM First Principles Study WIP (G1), macOS vsync + software-GL copy (C5 near-term). PR #2.
- **13 Aug 2026 — Phase 2.** B1 unified toolbar, B2 native chrome, B4 bundle id, C1 arm64 pin, C2 optional LTO/mcpu, C3 TBB.
- **13 Aug 2026 — Phase 1.** A1 SolidWorks nav defaults, A2/A3 pack + cube, A4 first-start apply.
- **Earlier — env-setup branch.** Original A/B/C scoping LaTeX and Cloud Agent environment notes. Not on this branch.

Effort ratings describe technical scope (breadth of components and invasiveness), not calendar estimates. “Exists today” items are configuration/defaults; “new work” items add code behind existing extension points. Compile with

```
cd Documents && ./build.sh.
```